10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Review Exercise 11 — Information Handling

Unit 11 | Review + MCQs | New Book (PECTAA)

Review Exercise 11 (Solved)

The review repeats the whole unit in a short way: one graph for the median, quartiles, deciles and percentiles, one box plot, one scatter diagram, and the range, variance and standard deviation of ungrouped and grouped data.

1. Four possible answers are given for the following questions. Choose the correct answer.

(i) _____ is used to get the cumulative frequencies.

(a) Addition (b) square root
(c) multiplication (d) division

Answer: (a) Addition — each frequency is added to the total of the frequencies above it

(ii) Second quartile represents:

(a) mean (b) mode
(c) median (d) variance

Answer: (c) median — Q_2 = median, the middle value of the data

(iii) First quartile divides the data into _____ equal parts.

(a) Two (b) Three
(c) four (d) ten

Answer: (c) four — quartiles cut the data into four equal parts

(iv) Difference between the highest and the lowest values is called:

(a) mean (b) variance
(c) range (d) standard deviation

Answer: (c) range — range = highest value − lowest value

(v) Scatter diagram represents a relationship between _____ variables.

(a) five (b) two
(c) three (d) four

Answer: (b) two — each point is one pair of two variables

(vi) _____ is measure of dispersion:

(a) mean (b) median
(c) mode (d) variance

Answer: (d) variance — mean, median and mode are measures of average, not of spread

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(vii) Positive square root of variance is called:

(a) mean (b) median
(c) standard deviation (d) range

Answer: (c) standard deviation — $\sigma = \sqrt{\sigma^2}$

(viii) _____ is not measure of dispersion.

(a) range (b) arithmetic mean
(c) variance (d) standard deviation

Answer: (b) arithmetic mean — the arithmetic mean is a measure of central tendency

(ix) Variance of the data 8, 8, 8, 8, 8, 8 is:

(a) 0 (b) 16
(c) 8 (d) 48

Answer: (a) 0 — every value is equal to the mean, so each deviation is zero

(x) Range of first 20 natural numbers is:

(a) 20 (b) 10
(c) 19 (d) 30

Answer: (c) 19 — range = 20 − 1 = 19

2. The following results for the long jump were recorded. Construct the cumulative frequency polygon and locate median, Q_1, Q_3, D_7, P_{90} and interquartile range on it.

Long jump results

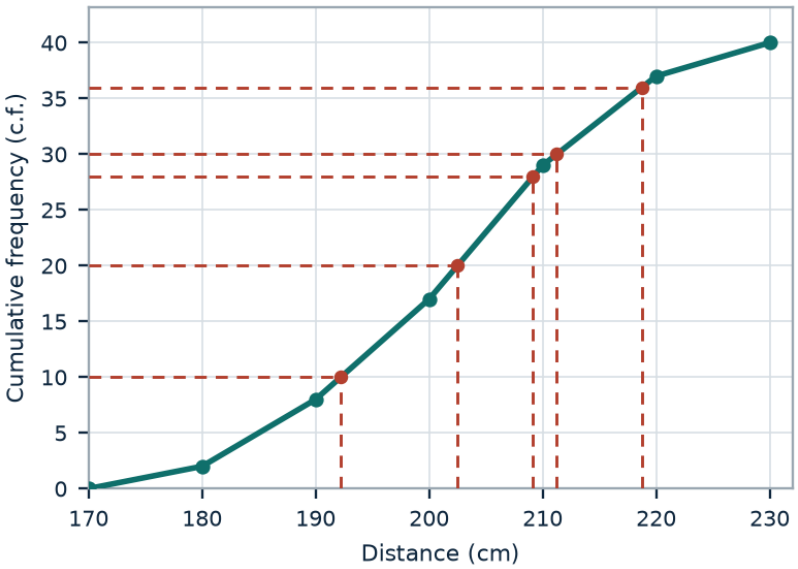
Distance (in cm)	170 – 180	180 – 190	190 – 200	200 – 210	210 – 220	220 – 230
f	2	6	9	12	8	3

Here $\Sigma f = 2 + 6 + 9 + 12 + 8 + 3 = 40$

Table of cumulative frequencies

Class	f	c.f.
170 – 180	2	2
180 – 190	6	8
190 – 200	9	17
200 – 210	12	29
210 – 220	8	37
220 – 230	3	40

Position of median = n / 2 = 40 / 2 = 20
Position of Q_1 = n / 4 = 40 / 4 = 10
Position of Q_3 = 3n / 4 = 3(40) / 4 = 30
Position of D_7 = 7n / 10 = 7(40) / 10 = 28
Position of P_{90} = 90n / 100 = 90(40) / 100 = 36



Cumulative frequency polygon and the readings taken on it

Readings taken from the ogive

Measure	Position on y-axis	Reading on x-axis
Median	20	202.5
Q_1	10	192.22
Q_3	30	211.25
D_7	28	209.17
P_{90}	36	218.75

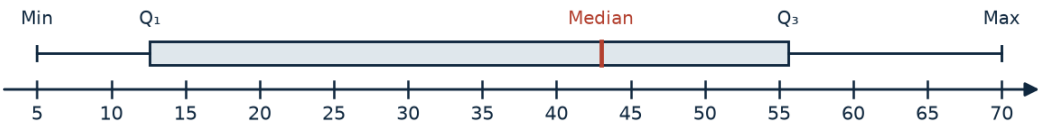
Interquartile range = $Q_3 - Q_1 = 211.25 - 192.22 = 19.03$

Answer: Median = 202.5, Q_1 = 192.22, Q_3 = 211.25, D_7 = 209.17, P_{90} = 218.75, IQR = 19.03

3. The summary statistics for a data set is given below. Show it with a box-and-whisker plot.

Summary statistics

Min. Value	Max. Value	Q_1	Median	Q_3
5	70	12.6	43	55.6



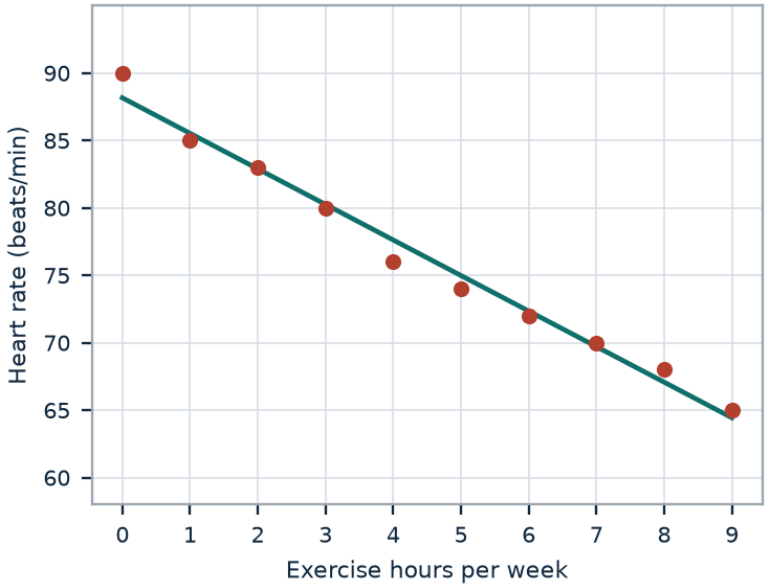
Box-and-whisker plot of the given summary statistics

Answer: Box plot drawn with Min 5, Q_1 12.6, Median 43, Q_3 55.6, Max 70

4. The following table shows the weekly hours spent exercising and resting heart rate (beats per minute) for ten individuals. (i) Plot the data on a scatter diagram and draw a line of best fit. (ii) State the type of correlation observed. (iii) Predict the heart rate of Sakeena who exercises for 6.5 hours per week.

Exercise hours and heart rate

Exercise hours	0	1	2	3	4	5	6	7	8	9
Heart rate	90	85	83	80	76	74	72	70	68	65



Scatter diagram with the line of best fit

The points fall from left to right, so the correlation is negative.

At x = 6.5 hours the line of best fit gives y ≈ 71 beats per minute.

Answer: (i) Graph drawn (ii) strong negative correlation (iii) about 71 beats per minute

5. Find the range for the given data: 25, 30, 35, 40, 50, 60, 65, 75.

Highest value H = 75, lowest value L = 25

Range = H − L = 75 − 25 = 50

Answer: Range = 50

6. Calculate range, variance and standard deviation for the following data set:

Grouped data

Class Interval	0 – 5	5 – 10	10 – 15	15 – 20	20 – 25	25 – 30
f	4	12	20	24	16	4

Range = H − L = 30 − 0 = 30

Here Σf = 4 + 12 + 20 + 24 + 16 + 4 = 80

Table for mean and variance

Class	f	x	fx	$x - \bar{x}$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
0 – 5	4	2.5	10	−13	169	676
5 – 10	12	7.5	90	−8	64	768
10 – 15	20	12.5	250	−3	9	180
15 – 20	24	17.5	420	2	4	96
20 – 25	16	22.5	360	7	49	784
25 – 30	4	27.5	110	12	144	576
Σ	80		1240	0		3080

Mean = $\Sigma fx / \Sigma f = 1240 / 80 = 15.5$

$\sigma^2 = \Sigma f(x - \bar{x})^2 / \Sigma f = 3080 / 80 = 38.5$

$\sigma = \sqrt{38.5} = 6.20$

Answer: Range = 30, variance = 38.5, standard deviation = 6.20

7. The sum of 5 numbers is 45 and the sum of their squares is 421. Find the mean and standard deviation of the data.

Mean = $\Sigma x / n = 45 / 5 = 9$

$\sigma^2 = \Sigma x^2 / n - (\Sigma x / n)^2 = 421 / 5 - (9)^2 = 84.2 - 81 = 3.2$

$\sigma = \sqrt{3.2} = 1.79$

Answer: Mean = 9, standard deviation = 1.79

8. The monthly household expenses (rupees in thousands) of two families for 6 months were — Family A: 45, 47, 46, 48, 46, 47; Family B: 38, 52, 40, 50, 42, 49. Calculate the mean and standard deviation of the monthly expenses. Which family spends more on average? Which family has more stable expenses?

Family A: $n = 6$

Mean = $(45 + 47 + 46 + 48 + 46 + 47) / 6 = 279 / 6 = 46.5$

Deviations: −1.5, 0.5, −0.5, 1.5, −0.5, 0.5

$\Sigma(x - \bar{x})^2 = 2.25 + 0.25 + 0.25 + 2.25 + 0.25 + 0.25 = 5.5$

$\sigma^2(A) = 5.5 / 6 = 0.917$, $\sigma(A) = \sqrt{0.917} = 0.96$

Family B: $n = 6$

Mean = $(38 + 52 + 40 + 50 + 42 + 49) / 6 = 271 / 6 = 45.17$

Deviations: −7.17, 6.83, −5.17, 4.83, −3.17, 3.83

$\Sigma(x - \bar{x})^2 = 51.41 + 46.65 + 26.73 + 23.33 + 10.05 + 14.67 = 172.84$

$\sigma^2(B) = 172.84 / 6 = 28.81$, $\sigma(B) = \sqrt{28.81} = 5.37$

Answer: A: mean = 46.5, σ = 0.96; B: mean = 45.17, σ = 5.37. Family A spends more on average and its expenses are far more stable

9. The daily wages of 40 workers in a factory are grouped as follows. Find the mean, variance and standard deviation of the daily wages.

Daily wages of 40 workers

Daily wages (Rs.)	1000 – 1200	1200 – 1400	1400 – 1600	1600 – 1800	1800 – 2000
No. of workers	6	10	12	8	4

Table for mean and variance

Class	f	x	fx	$x - \bar{x}$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
1000 – 1200	6	1100	6600	−370	136900	821400
1200 – 1400	10	1300	13000	−170	28900	289000
1400 – 1600	12	1500	18000	30	900	10800
1600 – 1800	8	1700	13600	230	52900	423200
1800 – 2000	4	1900	7600	430	184900	739600
Σ	40		58800	0		2284000

Mean = $\Sigma fx / \Sigma f = 58800 / 40 = 1470$

$\sigma^2 = \Sigma f(x - \bar{x})^2 / \Sigma f = 2284000 / 40 = 57100$

$\sigma = \sqrt{57100} = 238.96$

Answer: Mean = Rs. 1470, variance = 57100, standard deviation = Rs. 238.96

Quick Revision — Unit 11 at a Glance

c.f. table → ogive; median at n/2, quartiles at n/4 and 3n/4.

Deciles cut the data into 10 parts, percentiles into 100 parts.

Box plot shows Min, Q_1, Median, Q_3 and Max; IQR = $Q_3 - Q_1$.

Scatter diagram shows the correlation of two variables.

Range = H − L; σ^2 measures spread, σ is its square root.